

explain. write. edit. summarize: An Exploratory Study on Agency Negotiations in Student-Chatbot Conversations.

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Abstract

LLM-based chatbots such as ChatGPT have given technologies that traditionally operated on the back end a user-friendly, conversational interface. Their rapid adoption among students has prompted universities worldwide to issue guidelines and re-examine existing practices. Supplementing prior research based on discourse analysis and self-reported measures (e.g., surveys and interviews), we propose an approach for analyzing naturally occurring student–chatbot interactions, rooted in conversation analysis of *chats* ($n = 503$) donated by eight current Danish university students. The analysis identifies conversational patterns across three main types of activities and examines how agency is negotiated across the structural dimensions of signification, domination, and legitimation. Despite methodological limitations related to the sample, this study offers a promising path toward understanding how human–machine relations are recursively shaped in dialogue.

Keywords: LLMs, ChatGPT, communicative AI, agency negotiation, conversation analysis

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Introduction

OpenAI's launch of ChatGPT in November 2022 was a pivotal moment in the introduction of large language models (LLMs) as an everyday consumer product adopted by people for purposes across educational, work, and nonwork domains. The dialogue systems (or conversational agents) like ChatGPT are based on a versatile general pre-trained model that can support a series of tasks ranging from ideating, text generation, summarization, and translation. Compared to prior machine learning systems like personalized recommender systems, facial recognition, and predictive scoring algorithms which operate on the back end, OpenAI introduced a chat-based interface mediating interactions between the user and the underlying model in a natural language and dialogical form. Although chatbots (such as Replica) and virtual agents (such as Siri or Alexa) as a phenomenon and a commercial product have existed before 2022, ChatGPT combines a chat-based interface with an unprecedentedly powerful language model. The fast proliferation of LLM-based chatbots have sparked discussions about potential benefits related to productivity and automation weighed against challenges related to output quality and bias (Lucy & Bamman, 2021), environmental concerns (Rillig et al., 2023), predatory business models (Burkhardt & Rieder, 2024), and displacement of human agency (Zhao et al., 2024). The discussion has been particularly prevalent in education in which chatbots are both staged as a supportive partner or collaborator in the learning process (Mollick & Mollick, 2023) or as pedagogically perverse and unfit for educational purposes (Flenady & Sparrow, 2025). The current literature has largely been devoted to grasping the rapid adoption among students (e.g., Stöhr et al., 2024; Strzelecki, 2024) and institutional responses, for instance, formulated in university policies and guidelines (e.g., Driessens & Pischetola, 2024; Luo, 2024).

The aim of this article is to offer an alternative entry point to the discussions on the ramifications of LLM-based chatbots use in higher education and beyond. Instead of prescribing a universal form of agency to chatbots, we draw on the concept of *agency negotiations* (Gibbs et al., 2021) and suggest a methodological approach to capture and analyze how agency is negotiated in naturally-occurring dialogues between students and their chatbots. Demonstrating the framework, we conduct a pilot study entailing in-depth conversation analysis of ChatGPT conversations donated by eight Danish University students. The study explores *what characterizes conversations between Danish University students and chatbots and how is human-machine agency negotiated in chatbot dialogues?* The conversation analysis reveals preliminary insights into communicative patterns across central activities that AI-based chatbots are used for: to explain and summarize existing knowledge, generate and edit communicational products, and help and propose suggestions in personal and social situations. Our analysis informs a discussion on different ways machine agency is activated through *consultation*, *conversation*, and *registration* (Bordewijk & Van Kaam, 1986). Despite the limitations of the sample size, our approach offers a rich insight into how conversations between Danish students and a chatbot are structured, and how students negotiate agency in a dialogue with a chatbot. Our study contributes primarily to the research in the field of human-machine communication, and it can also assist educators in their practice.

Related Work

Agency Negotiation in Communicative AI

The field of human-machine communication (HMC) distinguishes itself from the broader field of communication by considering technology not only as a medium, but also, in certain situations, an active partner in communication with people and with other machines (Gunkel, 2012; Guzman & Lewis, 2020). Communication between humans and machines have long historical traces, but the advancements of large language models (LLM) and generative AI have turned communicative AI into an increasingly common everyday phenomenon (Simon et al., 2025).

Communicative AI is broadly defined by Guzman and Lewis (2020) as “technologies designed to carry out specific tasks within the communication process that were formerly associated with humans” (p. 72). LLM-based chatbots do that in several ways. Chatbots like ChatGPT, Gemini, and LaMDA rely on LLMs as their foundation model. The foundation model of LLMs consists of a large vector space connecting words trained on a massive quantity of digitized texts. Due to the vast quantity and variety of training data, LLMs can be considered as task-agnostic and unfinished (Burkhardt & Rieder, 2024), hence adaptable to a variety of purposes. LLMs are commonly altered by users through prompting in chat-based interfaces where knowledges stored in models are connected and made relevant. As a communication system, the chatbot interface supports an open-ended communication in which an output generated by the chatbot can be iterated with a new request, adjustment of the prompt, or inquiries into the output. The chatbot, in other words, can *talk back*, make unexpected connections, and their open-ended nature supports not only practical but also social functions in people’s everyday lives (Mahnke & Bagger, 2024). Such communicative patterns associated with LLM-based chatbots like ChatGPT support the appearance of what Natale and Depounti (2024) coin *artificial sociality*. This form of sociality is constituted by imitation of selected human communicational patterns, for instance dialogical forms, serving flow in the communication. At the same time, this does not mean that people are deceived to think that machine sociality is the same as the human one. Rather, it is argued, that because the communication is with a machine, the human can ascribe new functions to the relationship.

Considering machines as more active communicative partners also questions existing conceptions about agency of humans and machines. HMC scholarship has constructively pointed out that communicative AI systems act on multiple levels. Guzman & Lewis (2020), for instance, discuss how boundaries between human and machines blur at functional, relational, and metaphysical levels, and Hepp et al. (2023) argue for considering human-machine agency as an entangled relationship between human practices and technological infrastructures that

opens up a way of thinking about the agency of communicative AI at all levels of the social scale, combining the perspective of the individual and its interactional relations by understanding figurations such as organizations and communities as collective actors. (Hepp et al., 2023, p. 51)

A critical point that is transferrable to studies on human-chatbot conversations is that communicative AI technologies are simultaneously flexible tools, which can be prompted by the user to serve several functions, while also being a technical infrastructure and a communication system enabling, constraining, but also archiving communication (Jensen, 2025).

Agency at the interface level is, in other words, not the only situation for enactment of human-machine agency, but a critical moment where the role and function of chatbots are performed in interaction. Addressing this dimension of agency, we refer to *agency negotiations*. According to Gibbs et al. (2021), agency negotiations in HMC recognize that human and machine agency are recursively shaped in interactions on multiple levels. Machines do not possess intentionality or consciousness as people but machines can still enact agency: “Although intelligent technologies are not aware of the reasons behind their actions and cannot exercise discursive consciousness, they are capable of enacting more routinized practical consciousness that directs actions in a larger structure” (Gibbs et al., 2021, p. 158). This aligns with Natale and Depounti’s (2024) argument that, despite not having mental states, machines can enact sociality by imitating conversational patterns (including social norms) of human communication but also extends it by appointing that the actions performed through a communicative AI can, in turn, impact broader institutional structures. Drawing on Giddens (1984) structuration theory, Gibbs et al. (2021) further exemplify how the structure-agency in AI technologies is expressed through signification, domination, and legitimation (p. 160). Transferred to communicative AI, *signification* offers a lens to reflect on how meaning and agency are negotiated through discursive signals in prompts and replies; *domination* can be used to address how users and AI technology take turns controlling each other depending on the role that the user attributes to technology (e.g., consultant/collaborator); *legitimation* points to the institutional norms and values surrounding communicative AI and conditioning its use. Thus, despite the intent to origins within organizational contexts (Gibbs et al., 2021), agency negotiations offer useful frameworks to reflect on the functions and roles that people assign to chatbots in the dialogues (Banks & Graaf, 2020; Sundar, 2020). In agreement with Natale and Depounti, we think of chatbot’s agency as emerging in dialogue, and as a result of interactional behaviors.

Conversation Analysis

For the analysis of agency negotiations in naturalistic conversations between Danish university students and ChatGPT, we apply the analytical framework of conversation analysis (CA). Initially developed to analyze patterns of naturally-occurring human interactions (Garfinkel, 1967) and applied to study interpersonal telephone communication (Sacks et al., 1974), CA focuses not on the message, but on the structure of conversation: turn-taking, sequence organization, repair mechanisms, and nonverbalized cues reflecting ordering the dialogue. Rather than seeing structure as an “objective, external source of

constraint imposed on the individual” (Hutchby & Wooffitt, 2008, p. 4), CA regards structure as features actively made relevant, contested, repaired, and negotiated in interaction. This framework of analysis suits our goal of examining the structure and agency negotiation in dialogues naturally occurring between a human and a chatbot.

CA has been previously used in the field of computer-mediated communication (CMC) (Herring, 2010; Hutchby & Wooffitt, 2008; Meredith, 2019) to understand how the coherence of meanings is maintained and repaired, and how asynchronous conversations result in actions (see e.g., Paulus et al., 2016). Meredith discusses how turn-taking, sequence organization, and repair take place in online environments: The lack of embodied conduct is compensated with the use of smileys, emojis, and creative punctuation; there are varying lengths of gaps between turns; more room to self-correct before messages are sent; and there are opportunities for appending visual or textual elements to the conversation. Applying CA to studying conversations between humans and LLMs is, therefore, a natural next step. It was recently done in a special issue of *Discourse & Society* (Stokoe et al., 2024), for instance, in relation to action formulations (Liesenfeld & Dingemanse, 2024) and repair manifests in conversations (Pütz & Esposito, 2024). The key distinction here from other forms of communication is that such conversations happen not between two or more cognizant individuals, but a person and a model interfaced through a conversational system. Thus, for example, repair is not directed toward the meaning shared among the actors (Pütz & Esposito, 2024; Stokoe et al., 2024), but rather toward creating a satisfactory output.

Methodology

In the following we outline our process for data collection and analysis, as well as critical reflections on limitations of the study and ethical considerations.

Data Collection

The present study’s analysis is based on the data collected via a data donation approach, where participants donate for scholarly purposes *data download packages* (DDPs), which they collected themselves from digital platforms (Ohme & Araujo, 2022). This methodological approach provides a rich and unobtrusive insight into users’ behaviors, which mirrors the data available to platform companies. The choice of ChatGPT as an example of communicative AI and an LLM is justified by it being by far the most frequently used chat interface in Denmark (Simon et al., 2025) and the first tool to reach widespread adoption. Conversations with ChatGPT has a higher chance of being older (since its launch in 2022), more sophisticated, as users have had enough time to familiarize themselves with the technology. The DDPs of ChatGPT included prompts and answers, timestamps, conversation headlines, and the model’s name.

TABLE 1
Overview of Participants and Interactions

ID	Discipline	Years of enrollment	Number of conversations	Number of prompts by user	Average number of characters per prompt	Number of days between first and last use
P1	Architecture	4	67	194	2192	228
P2	Biology	3	28	245	250	467
P3	Digital media	2	113	498	255	277
P4	Digital media	0	186	753	953	731
P5	Media	1	41	302	2834	273
P6	Information	2	40	161	52	423
P7	Journalism	3	12	44	371	321
P8	Social science	1	16	207	974	228
Total		2*	503	2404	1004*	368.5*

Participants for the study were recruited the following way. First, the protocol for data gathering was pre-approved by the Institutional Ethical Council and GDPR-registered according to the university protocols. Second, students were freely approached by the authors in classrooms and in university halls and cafés. They received information (in oral and written form) about the goals of the study (analysis of their conversations with the chatbot), instruction guide for acquiring and donating the data, and details about data handling, including practices for dissemination, anonymization, and security. The participants were also encouraged to delete conversations that they considered sensitive before donating their chat logs. Since the process of donation was fully in the students' hands and conducted in their free time, they could easily reject participation without being pressed to it on the spot. The authors have informed each student that excerpts from their logs may be used anonymously as illustrations in academic publications. Finally, DDPs were received to an email address, specifically created for that purpose, and later were gathered in a folder on the university's server.

From approximately 200 students that were approached by the authors, eight DDPs were collected, containing in total 503 single conversation between a student and ChatGPT. At the time when the data was gathered, the recruited students had studied between 0 and 4 years at the university (avg = 2). With Danish as their primary language, they all have good conduct of English, which is taught at a high level in Danish schools (and instruction is partially done in English). Our data demonstrates that the students who donated their ChatGPT logs have been using the LLM-based chatbot for a few years. Although we

expected a somewhat larger donation size, this data composition allowed us to go more in-depth into each conversation using CA as the analytical lens.

Data Analysis

Each of the eight DDPs consists of one JSON and one HTML file containing textually recorded conversations and associated metadata (e.g., timestamps and model name). HTML files record conversations in a user-friendly way, just as they appear in the ChatGPT interface. During the analysis, the JSON file was used to gain quantitative overview of the data (see Table 1), whereas the HTML file was used in the qualitative analysis of the conversations. The HTML files were printed out and cut into the units of the analysis—single conversations, which were then sorted and categorized roughly into three major themes according to the primary activity the conversation supports. The activities include activities that instruct the chatbot to *explain and summarize* existing knowledge or information, *generate and edit* new information and communicational products, and help and suggest with personal or social matters. The three themes are not exhaustive and contain overlaps, as each conversation may contain several themes and flow from one to another. Thus, the divide mainly serves the purpose of facilitation of the further CA.

The three activities were analyzed independently for conversational patterns using selected concepts within CA (Meredith, 2019). As formerly mentioned, this included translation due to the nature of HMC. Specifically, we analyzed *opening sequences* of the user, including commands and requests (e.g., action verb), as well as the context of the conversation (context descriptions, attachments, etc.). Opening sequences offer important information about purposes and functions of conversations; for example, the topicality and open-endedness of the conversation. *Sequence organization* was used to probe into the dialogical structure of the conversation. While conversations follow a formalized turn-taking order (the user prompts—the chatbot responds), sequences may have different numbers of turn-taking pairs, lengths, topics, and foci (sometimes topics change within one conversational sequence), as well as uploaded documents and contextual information. Also, we analyzed *repair* in conversations, focusing mainly on the ways that users adjust their prompts in response to the output, but also on repair actions initiated by the chatbot when it makes a request for more details and additional context. Finally, we adapted the concept of *embodied cues* to capture *textual cues* in the conversation. While smileys and emojis can convey meaning in text-based conversations, we find that ChatGPT emphasizes concepts and arguments rather through formatting: headings, bulleted or numbered lists, putting some text in bold. Moreover, with reference to textual cues, we also analyzed the tone of conversations.

Methodological Limitations and Ethical Considerations

Before continuing to the results, it is worth noting that the results must be read in context of the exploratory nature of this research, and ethical considerations connected with data donations of ChatGPT conversation. The obvious limitation of this study is its sample size and quality. The eight students were recruited by a combination of convenience and snowball sampling approach, resulting in a sample of Danish students with variations in years of enrollment and academic discipline, as well as uneven frequency of ChatGPT usage. The

sample is by no means fit to make generalizable claims about chatbot use among Danish university students. However, the 503 donated conversations analyzed offer a rich first entry into the integration of ChatGPT in everyday academic tasks of reading and writing, as well as an overview of the subtle ways that roles are assigned in interaction.

The low participation rate was disappointing but not unforeseen, as sampling is a common challenge in data donation studies, given the technical complexity to participate and sensitive nature of the data material (Strycharz et al., 2024; Van Driel et al., 2022). Moreover, gathering donations when policies about AI use at Danish universities were unsettled is likely a factor that has hindered participation and an ethical consideration. When preparing the data collection, we recognized that our data material most likely contained personal information and potentially data traces that could indicate misconduct or deviation from university policies. To avoid harm for participants and ensure their privacy, we emphasized the possibility of deleting data parts before submitting the donation, minimized data stored to include the raw conversations and metadata such as timestamps, titles of the conversation, model details, and IDs for the conversation and prompt. Also, data access was reduced to the named participating researchers, and the data was stored on the university secure drive for a period of 5 years. The raw conversations, which sometimes contained names and locations, were not edited after submission and left there as is. The protocol was registered in the GDPR archive and pre-approved. The protocol was reviewed and approved by the faculty ethical committee.

Agency Negotiations in Student-Chatbot Interaction

The analyzed conversations reveal that chatbot interaction is a versatile form of communication which can be used to seek and summarize information, generate and edit text, support project work, and offer personal help and suggestions. Chatbot conversations vary in their length, structure, level of detail provided in a prompt, social and emotional cues, and the agency attributed to the machine. The following presents the conversation analysis within the three identified activities and contextualizes the findings in relation to agency negotiations.

explain. summarize.

In our data, students frequently use ChatGPT to support their learning, including such activities as information seeking, summarizing texts, and explanation of concepts. The turn-taking structure of these conversations varies in length from question-to-answer queries to longer interactions with follow-up questions, clarifications, and requests for translation.

When using ChatGPT in search of information, students query the model and its memory in a consultation-like pattern. The students tend to formulate the request as they would in a conversation with a human, as a *wh*-type question: “what is long form journalism” (P5) or “why do people move to the suburbs” (P1). Another common way is to utilize keywords, similar to a modern internet search: “tell [me about] spanish politics and culture” or “please describe to me what it requires to have a good UI/UX design.” In our sample of conversations, we see a pattern of information search less dominated by the

use of keywords and more benefitting from the advantage of the large semantic context that LLMs offer (Kocoń et al., 2023). Thus, the information search appears to be more intuitive, with short question–answer sequences and only a few turns. One student (P5) asks the chat for help remembering a film that they saw in childhood, appealing to a more associative type of reasoning rather than a strict rule-based information search typical of an archive or a library.

Table 2 illustrates a common query sequence used for the purpose of understanding, which includes (with timestamps) the structure of an opening sequence, ChatGPT’s answer, user’s feedback and confirmation, and a closing. The student (P2) aims to clarify the meaning of some information they read in a research article. The chatbot’s response resembles the tone of the opening sequence suggested by the student (e.g., “you’re”, “let’s”, and use of exclamation marks). Moreover, the formatting of the response includes listing of points in bullets and highlighting keywords in bold, supporting quick reading. This brief interaction demonstrates how the chatbot not only supplies information but also supports interpretation and evaluation of information.

TABLE 2
Example of Information-Seeking

Actor	Timestamp (hh:mm:ss)	Text	Annotations
P2	08:00:41	Good morning! I am reading an article where they see that <i>Enterococcus faecalis</i> became resistant to its specific phage VD13 already after 2 days (28% were resistant) and after 10 days 68% were resistant. Would you say that is a lot, and/or fast?	Greetings Context Request
ChatGPT	08:00:56	Good morning! Yes, the rapid development of resistance to phages by <i>Enterococcus faecalis</i> that you’re describing is quite striking, both in terms of speed and the high percentage of resistant cells. Let’s break it down: [Three points are made about speed, percentage, and comparison to other systems]. Summary: Yes, this is quite a fast and significant development of resistance. It highlights the dynamic interaction between phages and bacteria and shows how quickly bacteria can adapt to phage predation when under selective pressure.	Greetings Short reply Three bullets for detail Summary Formatting: Bullets, bold text
P2	08:02:03	Thank you so much, that was very helpful	Feedback/confirmation
ChatGPT	08:02:05	You’re very welcome! I’m glad you found the explanation helpful. If you have any more questions about phages, bacteria, or anything else, feel free to ask!	Closing

Another common structure of the query that students apply in our sample involves embedding an object (e.g., a body of text) as an input and sometimes as a conversation opener for the purpose of summarization and/or clarification of concepts/ideas, thus providing the context and guiding the conversation. Such queries tend to entail a larger number of turns where the initial summary is repaired and extended. Table 3 illustrates a conversation sequence, where the student (P2) requests a summary of the included text, which is then condensed by ChatGPT from 1,300 to 130 words, including a summary of the key points. The student then asks for a bullet-point summary, and a translation to Danish (for better comprehension and potentially direct use in one of the school tasks). The sequence takes about 2 minutes until the next passage of text is added for the chatbot to summarize in the same format with instruction like “SHORTEN the text, thank you.”

TABLE 3
Example of Text Summarization

Actor	Timestamp (hh:mm:ss)	Text	Annotations
P2	21:28:45	give a resume of this text [1,300-word academic text added]	Request Context (text)
ChatGPT	21:28:49	Resume The text explores the systematic measurement of behavioral effects in [. . . 130-word resume]	130-word resume Formatting bold
P2	21:29:25	What are some key points?	Repair of the resume
ChatGPT	21:29:33	Some key points include: • Scarcity of Behavioral Studies: Historically, there has been a focus on analyzing attitudes and perceptions in communication research, with fewer studies systematically examining behavioral effects. • [.. another seven summarizing points are made] Overall, the text underscores the significance of systematically measuring behavioral effects in communication research for a comprehensive understanding of media effects on individuals.	Formatting of summary in numbered list of points Meta-text before and after points
P2	21:30:03	in danish	Repair (language)
ChatGPT	21:30:17	[eight points are translated to Danish]	Repair
[. . .]	21:32:06— 21:45:00	The conversation iterates between user adding text and requesting summaries of passages and the chatbot replying with listed summaries with few repair actions such as “SHORTEN the text, thank you”	

Within the theme of understanding, it is evident that agency between students and ChatGPT is negotiated on several levels (Gibbs et al., 2021). At the system level, students exert a form of *domination* where the chatbot is treated as a subordinate social actor. The chatbot is designed to never initiate a conversation, to always reply, and then to wait for the user's response. The student, thus, is in control of the topic and sets the tone of each conversation (formal/nonformal, playful, etc.), connecting to the *signification* dimension of agency negotiation. At the same time, the chatbot enacts agency through its algorithmic procedures of recalling knowledge stored in the foundation model (Burkhardt & Rieder, 2024): answers to questions, summaries and choices of key points, as well as the quality of translation. As the previous example demonstrated (Table 3), the chatbot can render complex and long texts into dense bullets. Differing from information seeking, the summarized text is appended, thus available to the user, which supports double-checking and monitoring output quality. Finally, by requesting new clarifications, summaries, and formatting, the student negotiates control (Gibbs et al., 2021) over the output of the conversation, trying to bring it closer to the expected outcome. For instance, one student (P5) uses the chatbot to summarize a text about language and semiotics, asking follow-up questions like “what characterizes a constructivist approach” and “what are iconic and index signs?”

generate. edit.

Where *explain and summarize* refers to conversations that support users' need for understanding recorded knowledge, *generation and editing* refers to conversations in which the chatbot supports creating or editing ideas or communicational products that can be transferred into and make a difference in the world. The requests for content generation can be identified in the opening passages of conversations by such call-to-action verbs as *generate*, *write*, *make suggestions for*, *come up with a*, *rewrite*, *edit*, *shorten*, followed by a description of an idea and contextual information. The latter varies in length and depth, depending on the type of task requested from the chatbot (e.g., develop a first draft, suggest ideas, edit the existing text, or debug a piece of code). A common request among the students in our sample is to generate text in a certain format and tone, with such applications as formal letter writing, rejection messages, requests for interviews, or motivation/cover letters for jobs and exchanges. For example, one student (P1) requests “write a motivation letter as part of a student visa” and offers details about the country, university, courses, and prior experiences and affiliations to the country as contextual information. After the initial version of the letter suggested by the chatbot, the student makes a request for editing: “make it shorter,” followed by a description of one of the courses in more detail. Similar to the generated summaries discussed in the former section, the instructions to generate content tend to be instrumental orders, which later—in the editing phase of the conversation—can become more collaborative and equal patterns of communication.

Among the editing requests are proofreading and grammar checking, spelling, revising or changing style and tone, making an argument sharper, making the text more entertaining, and even finding poetic rhymes. A common command to the chat is simply “make this text better” or “rewrite the text,” which indicates the lack of need to extensively instruct the machine, as this technology allows for a more intuitive approach. There is a shift in the roles between the student and the chatbot here: from the user repairing the output of the

chatbot to the chatbot correcting the user’s product. In conversations, where text editing is the main goal, smaller portions of larger texts are edited sequentially (for example, in an academic report). It is mostly an instrumental form of exchange where the chatbot outputs the transformed text without providing any comments on the changes made.

Among the more creative applications within this function are idea generation and support in project development. The complexity of ideas ranges from suggestions of names for a company, possible activities with friends, to academic projects. In a few cases, the chatbot takes the role of a partner over an extended period of time, such as during a full DnD-campaign (P4) or as a collaborator on a student project. Table 4 illustrates a case of a complex collaboration where a student (P8) has a 62-turn long conversation with the chatbot over a 3-day period, where the chatbot assists in a development of a group project.

TABLE 4
Example of Collaboration

Actor	Timestamp	Text	Annotations
P8	23:04:10	These are the notes from our project: [two research questions, and four subquestions are] ... provide different responses to these three: Find empirical material that support the problem statements Look at theoretical positions to support the problem statements Look at collection of data and methods Respond to the above problem statements individually	Opening Context Request Request: Format
ChatGPT	23:04:22	User’s project focuses on two main problematics: [rephrasing of the problem statements]	Response
P8	23:04:22	Model context updated (meaning the user regenerates response)	Repair
ChatGPT	23:04:22	Here are some suggestions for theoretical, empirical data and methods which can support the problem statements individually: • Problem statement 1: [suggest theories, methods, and data material for each problem statement] These theories and methods will help you supporting the two-problem statement with relevant literature and empirical data.	Response Formatting: List + bold

Actor	Timestamp	Text	Annotations
P8	23:41:05	What is best to make document analysis on?	Request
..		[...]	
P8	23:41:24	What about reports that contain interviews?	Request
ChatGPT		Reports that contain interviews are an excellent source for document analysis especially if you want to analyze empirical data or qualitative descriptions from different actors	Confirmation

In this example, the chatbot assists in suggesting and helping with the choice of theory, methods, and empirical material that would *support their argument*. It both generates responses and offers opportunities to elaborate on its own suggestions, such as expressed in the following prompts: “what is best to do document analysis on?”, “what about reports that contain interviews?”, “how can one use Anthony Giddens for this?”, “what relevance does Foucault have within constructivism?”, and “but does he relate to constructivism at all?” In this example, the conversation goes beyond a simple consultation, as it resembles a form of brainstorming, pitching ideas, and making decisions together.

When it comes to negotiating agency (Gibbs et al., 2021), as a project partner, the chatbot tends to *legitimize* ideas by enriching and motivating them, while refraining from challenging the students’ thoughts. For example, doing document analysis on *documents containing interviews* could be considered as a rather unconventional approach to document analysis whereas ChatGPT describes it as an “Excellent source for document analysis” followed by a list of potential sources for analysis. More broadly, the generative function of LLM-based chatbots can be considered as a resource for people to act in the world by drafting and editing products spanning from academic assignments to computer code, and job applications. One of the students even employ a form of prompt engineering practices repeating the same prompt but changing the context indicating automation of a particular task (job application). In such situation, the *practical consciousness* associated with typing on a keyboard, selecting appropriate words, and structuring sentences is allocated to the chatbots. Students might indeed reflexively monitor the output and its appropriateness, but the redistribution can arguably both have implication on individual and institutional levels: On the individual level, this form of automation can be considered as a form of de-skilling (Zhao et al., 2024) where central linguistic capabilities are distributed from the human to the machine. Following Gibbs et al. (2021), the allocation of routinized practical actions can also affect larger institutional structures that, in context of text generation, include norms and values associated with creating, authoring, and evaluating written products (Gunkel, 2024). One student even appears to be aware of this breach with institutional rules and norms asking the chatbot to rewrite a sentence “so that it won’t be flagged for plagiarism.” Such a direct acknowledgment of an unethical practice is not followed by any judgment or even a recognition from the chatbot, which is always “happy to

help” and simply satisfies the user’s request. This example demonstrates how the student can exercise agency over the content without taking any responsibility for its quality.

help. suggest.

Personal and social interactions draw on the already described practices like information-seeking, idea generation, and text editing. Instead, the motives of conversations are connected to learning about oneself, reflecting on social dilemmas, and mediating activities and interactions with others.

Personal and social interactions appear less frequently in our data, and not all participants had them. Nevertheless, such conversations provide valuable insights into how chatbots are used with social intentions to seek personal advice, gain motivation, plan personal or social activities, and facilitate communication with others. Many of the personal and social interactions are simple requests for ideas on meal plans, creative hobby projects, presents, social activities with friends, rephrasing a difficult email, or writing a greetings card. These simple cases often entail few turns and vary by the more detailed information about the context of use.

The more interesting cases concern the situations in which users playfully engage with the chatbot for entertainment purposes or reflect on personal aspects and social situations. Playful uses include using the chatbot to *roast my car*, generating quizzes, or making small games like a gauntlet. Table 5 illustrates a personality test, which sorts the user into one of four Hogwarts houses. In this example, the student (P3) requests the chat to switch roles and ask the questions instead. A similar format is then repeated in a longer-form personality and relationship test of a deeper self-reflective character. In such tests, the student is addressed by name, compliments the described actions, and appraises for the ability to reply well to responses. The conversation, in other words, is a form of game for the student to see how the chatbot interprets them through their inputs. This boundary-pushing format of the ChatGPT use is rather rare, but it shows a broader range of application of this technology, where there is potential for more complex conversations, playful delegation of control to the algorithm, and a fun way to explore its limits.

In other words, the personal and social uses accounted here do not indicate a deception of ChatGPT as a friend, but rather that the system attributed social functions. As Natale and Depounti (2024) argue, sociality emerges not so much due to the systems having mental states and being social, but rather because they are programmed to imitate social behaviors. The analyzed conversations show examples of users learning the limitations of the model by exploring its boundaries; however, there are also several examples of users that share intimate details, including personal thoughts, family diseases, and social challenges. Users can share their thoughts in seemingly honest ways without risking any social sanctions. The chatbot can support the user by providing information, making an analysis of the situation, and offering advice—always in a constructive and encouraging tone. From the users’ perspective, such conversations legitimize their opinions and beliefs, whereas from a systemic perspective, intimate and personal information is fed into the system exposed for exploitation.

TABLE 5
Example of Personal Communication

Actor	Timestamp	Text	Annotations
P3	10:41:04	You have to ask me some questions about myself, to determine whether I am a gryffindor, hufflepuff, ravenclaw or slytherin	Request Context
ChatGPT	10:41:17	Sure, I'd be happy to help you figure out which Hogwarts house you belong to! Let's start with some questions about your personality, preferences, and values. • What do you value most? [. . . seven additional bullets] Feel free to answer these questions, and I'll help you determine which house might be the best fit for you!	Response Format: Listed, bold Request to user Summary
P3	10:44:40	1:B [responds to remaining questions in same form]	Response to chatbot
ChatGPT	10:44:54	Great, thanks for your answers! Let's analyze them to see which Hogwarts house you might belong to. [Bullet list of values assigned to Hogwarts Houses, assigns values to responses, and provides a characteristic] Does this sound like it reflects your personality?	Context Answer: Result of quiz Follow-up question

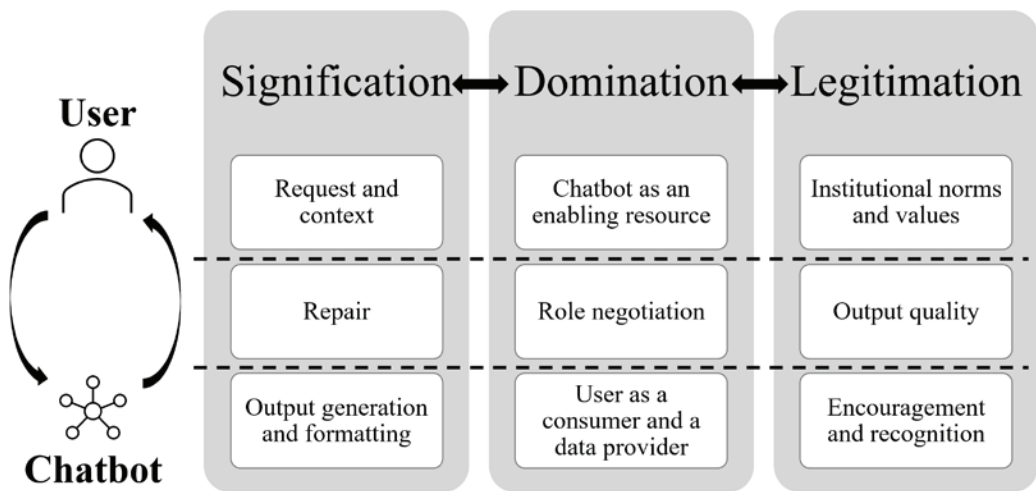
Summary

Our analysis indicates several ways that agency is negotiated in dialogue in conversations between students and ChatGPT. Figure 1 summarizes the findings mapped into the concepts of domination, signification, and legitimation (Gibbs et al., 2021; Giddens, 1984) on the horizontal axis, and whereas the vertical axis maps whether agency primarily is enacted by the user (upper), chatbot (lower), or negotiated by both actors (middle).

At the *signification* dimension, the conversations can be considered as co-constructed that require a user-defined request and context, and a chatbot's response. The interpretive flexibility of the chatbot is conditioned by the specificity of the opening request and contextual details supplied. Depending on the request, the chatbot can assert discursive agency ranging from simple tasks like translate or proofread passages, to more complex linguistic tasks like summarizing key points from long passages of text or generating initial drafts. In between these are repair actions: user-initiated repair specifying the intended outcome or repair initiated by the chatbot via editing users' texts or requesting additional

information to improve outputs. Repair is a particularly interesting part of conversations because it reveals frictions and misalignment between users’ expectations and the chatbot’s output disclosing to the user the limits and boundaries of the system for the intended task; for example, when a student uses the chatbot to remember a book or film that they saw in childhood. The chat runs out of plausible suggestions in the first few rounds, while the student continues the interrogation, forcing the chat to suggest increasingly more irrelevant options. At some point, the chatbot starts producing *hallucinations*, unable to admit ignorance.

FIGURE 1
Chatbot Use in Relation to Agency Negotiation



Domination recognizes that chatbots are resources that can enable certain actions, be it to efficiently understand existing knowledges, create texts, or self-reflect. While presented in our analysis as distinct activities, the analyzed conversation reflects that the students use ChatGPT for more activities, and that shifts in activities also frequently happen within conversations. In such situations, the chatbot can move from assuming the mere instrumental function (e.g., summarizing a text), toward being afforded a more collaborative role (e.g., seeking additional information or writing a paragraph). In return, users pay—either by subscription or through providing their personal data. Our results attest that users, beyond the mere instructions, both supply copyrighted information and deeply personal information which potentially serve the purpose of improvement of services or personalized advertisement.

Finally, *legitimation* recognizes that chatbot use is governed by institutional norms and values. The results show that the use of chatbots by students interferes with several institutional domains, including academia, job market, and interpersonal relationships. Legitimacy of ChatGPT usage is indeed conditioned by institutionalized norms, like university AI policies, but our data attests that these can intentionally be circumvented in use which in turn challenge existing conventions for text production, but also interpretation

and assessment. ChatGPT takes on most tasks requested by the user, formats replies in a reader-friendly manner, and encourages users to return. Also, ChatGPT does not push back or critique ideas and perspectives. This tendency to overcompliance is reflected in the chatbot apologizing *for the confusion and oversight*, as well as expressing support and reassurance, never confronting or criticizing the user. That is especially concerning in more boundary-pushing requests like in the case of editing somebody else's work to avoid being caught for plagiarism. However, the extent to which ChatGPT is used expectedly is connected to the users' judgment of whether the output quality suffices the task at hand. Such cases of repair and agency negotiation raise the questions of responsibility for the incorrect information, unoriginal work, and even more so for the potential effects on the quality of interpersonal relationships and learning.

Moving Between Conversation and Consultation

In this article, we explore *what characterizes conversations between Danish University students and chatbots and how is human-machine agency negotiated in chatbot dialogues?* We approach this question using a data donation methodology, tapping into conversations as they naturally occur. While our sample is limited ($n = 8$), the conversational logs demonstrate great richness in depth and variety and offer a promising path for future HMC research. However, it must also be recognized that this article only offers one perspective on agency which is connected to the materiality of LLM-based chatbots that operate on at least two levels: (1) At the training stage, diverse kinds of communication materialized in books, weblogs, online fora, and news articles just to mention a few sources that is processed, congregated, and stored in a model which users, (2) at the interaction stage can converse with. This is a curious form of many-to-one communication (Jensen & Helles, 2017) where *many* authors and texts are congregated into new texts, constructed based on users' information needs as they are expressed in their prompts. This fragments the communication process in time and challenges studies of human-chatbot interaction as a holistic process. In this study, we focus on the communication at the interface level (interaction stage), disclosing important details about students' actual uses, traits of the communication, and how agency is negotiated in interaction.

Our empirical material illustrates that ChatGPT is indeed a "Jack of all traits" (Kocóń et al., 2023, p. 9) assisting in various processes in students' lives, including seeking information, generating ideas, interpreting academic texts, writing assignments or job applications, planning activities, formulating personal messages, and reflecting on personal issues. ChatGPT is always available, and there seems to be no social consequences of disclosing inappropriate intentions, asking poor questions, misspelling words, and so forth. Rather, the chatbot tends to encourage users' intentions and mirror their tone of communication (Bender et al., 2021). The fragmented communication process is also prevalent in the dialogues.

ChatGPT is most frequently used for *consultation* in a similar manner as a search engine (Bordewijk & Van Kaam, 1986). This is particularly prevalent in uses that regard information-seeking, text generation, or rewriting of specific texts or messages. In such conversations, the tone is instrumental and instructive, and the structure of the conversation

entails few turns. The chatbot responds in a formal tone, for instance, by repeating the intention of task, listing arguments, and summarizing them in a conclusion. From an agency perspective this can, on the one hand, be considered a way to outsource core academic tasks, like academic reading and writing, to the machine. On the other hand, as a communication process, the user is in control of initiating and determining the output of the exchange (Bordewijk & Van Kaam, 1986).

In contrast to consultation, ChatGPT masks the communication as a *conversation*. The conversational traits are prevalent in interactions where students and the chatbot more collaboratively edit and refine texts, develop project ideas, and reflect on personal or social issues. When conversations are more dialogical, the language can be more ambiguous, affording exploration but also requiring more repair along the process. For users, this can mean more iterations to instruct the chatbot to deliver the anticipated output, whereas for the chatbot, it can entail requesting additional contextual details. Additionally, in interaction, students can utilize the capabilities of language models to a greater extent by combining, for instance, text summarization with the seeking of additional information or the generation of relevant examples. As a communication process, *conversations* are not as controlled by the user as consultations (Bordewijk & Van Kaam, 1986), exchanging the automation of specific academic tasks with a closer form of collaboration between the student and chatbot. The collaborative uses might afford users the opportunity to explore ideas and test the limits of the models, thereby gaining a deeper understanding of the underlying model.

On a final note, Bordewijk and Van Kaam (1986), also point out to *registration* as a form of communication in which the center of communication consults their users. Not as a communication partner, but a communication platform, ChatGPT registers the existing communication that feeds back to the system as meta-communication (Jensen, 2025). The gathered conversations reveal that some users share personal information about interests, habits, personal challenges, family illnesses, as well as potential future purchases, like a birthday present for a family member. Such information can be used for model training or commodification. While users by consultation and conversation can exploit the chatbot's ability and be influenced by its outputs, the capture of metadata reflects an asymmetry in the interaction.

Concluding and Future Work

As communicative AI expands and evolves as a societal phenomenon, it becomes increasingly important to understand how communication unfolds, stabilizes, and evolves in interaction. LLM-based chatbots—a prototypical form of communicative AI—we argue, is a communication process that is scattered in time due to the temporal differences between model training and the situation of use. Although we sympathize with the intention to examine and understand communicative AI across analytical levels (Hepp et al., 2023), the fragmented communication process presents challenges to doing this empirically. To strengthen the empirical foundations of communicative AI, an alternative approach is also about detailing what happens at distinct parts of the communication process. Our study offers a path to study the middle of this equation, namely the messages, based on a

data donations methodology (Ohme & Araujo, 2022). It points to diverse ways students use ChatGPT for academic and social purposes, and how agency is negotiated and allocated in the interaction. Data donations from ChatGPT are a naturalistic and unobtrusive source of information that can capture rich insights into the adoption of communicative AI, characterizing traits of the communication, and how it evolves over time—an insight usually reserved for large online platforms (Flensburg et al., 2025). Our limited sample of eight Danish university students is a symptom of the personal and technical barriers in the recruitment when it concerns donating personal and potentially sensitive information (Van Driel et al., 2022). This pilot study demonstrates how data donations can inform questions about agency negotiations in interaction which future studies are encouraged to expand on. The promising paths of research include to improve the sample of chatbot data donations, unlocking questions about the adoption of communicative AI, identifying patterns in the human-chatbot communication, and their development over time. Another promising path entails employing data donation as an elicitation method to provide deeper insights into how the outputs of chatbot conversations transfer to real-world situations. Also, deeper investigations into the role of repair in student-chatbot communication can shed light both on the functionality of this technology, its possibilities and limitations, as well as on the variety of tactics of navigating these limits that students apply in order to get the desired outcome.

As of today, the use of LLMs in schools and universities is still negotiated at the local and national level in Denmark. With the introduction of specific policies from the government and universities, it will become more pressing to study the student practices of generative AI use, and the ways that they impact learning. Today we are looking at the phase of technology adoption and exploration.

CRedit Authorship Contribution Statement

Árni Einarsson: Conceptualization, Methodology—research design, data collection, and analysis, Writing—original draft, Writing—review & editing, Project administration.
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